Sujal Almeida

+17744780007 | almeida.su@northeastern.edu | Boston, USA | Linkedin | Portfolio | Github

SUMMARY

Computer science graduate student at Northeastern University with hands-on experience in machine learning, privacy-preserving systems, computer vision, and AI automation. Experienced with Python, PyTorch, scikit-learn, and data analytics, with a focus on AI, ML, and security

EDUCATION

Northeastern University, Boston, MA
MS in Computer Science | 09/2026 - 12/2028
Coursework: Programming Design Paradigms, Cybersecurity

Fr. Conceicao Rodrigues College of Engineering, Mumbai, India
B.E. in Computer Engineering | 11/2022 - 05/2026
CGPI : 9.1/10
Coursework: Object Oriented Programming, Data Structures, Databases, Software Engineering

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++
Machine Learning & AI: PyTorch, scikit-learn, OpenCV, NumPy, Flower, LangChain, Gemini, Ollama
Privacy & Security: Federated Learning, Homomorphic Encryption, TenSEAL, CKKS
Web & Backend: Django, Flask, React.js, React Native

EXPERIENCE

PwC India — *02/2026 - 08/2026*

Advisory Intern-IDAC — *Mumbai, India*

- Developed **two AI/agentic proof of concepts**, including an LLM-based prompt amplifier for automated prompt enhancement.
- Built an AI agent for dashboard metric investigation, utilizing SQL to trace metrics to underlying data sources.
- Automated client file-upload reminders, reducing manual follow-ups across **75+** client interactions.
- Performed **SQL-based LDM and data-mart testing** using joins and cross-table validation.

ICICI Prudential Life Insurance — *12/2024 – 01/2025*

Machine Learning Intern — *Mumbai, India*

- Processed 1M+ data points, improving data quality by 30%.
- Engineered 25+ features, boosting model accuracy by 15%.
- Optimized ML models, increasing recall to 87%.
- Conducted post-model analysis for decision-making.

TECHNICAL PROJECTS

PrivMed | Privacy-Preserving Federated Learning Healthcare Platform — *07/2025 – 04/2026*

Technologies: Python, React, Flask, Flower, CKKS

- Developed a **privacy-preserving federated learning system** for distributed disease prediction.
- Applied **CKKS homomorphic encryption** to protect model updates during aggregation.
- Implemented encrypted model-update aggregation while keeping sensitive training data decentralized.
- Created a **React dashboard** for laboratory and administrative users.

CivicEcho | Crowdsourced Civic Issue Reporting and Resolution System — *09/2025 – 12/2025*

Technologies: React Native, Flask, MediaPipe, Hugging Face, Whisper, Vakyansh, Gemini

- Used multimodal analysis (text, image, voice) for rich incident reporting.
- Automated complaint triage and routing via agent-based orchestration.
- Integrated crowd and AI verification to filter spam and group duplicates.

PUBLICATIONS

AI with Agency, Cities with Resilience: Deploying Agentic Systems for Urban Disaster Response. Link
International Conference on Advancing Technology in Engineering and Science

Harnessing Generative AI for Precision Chemistry Link
International Journal of Computer Applications